



Doctor in astrophysics and software engineer, specialized in numerical simulation and performance optimization. Lead developer of simulation and data reconstruction pipelines for the QUBIC experiment, deployed on HPC clusters. Seeking a position in numerical simulation or HPC.

## PROFIL

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Permis B/A2

[TomLaclavère](#)

[Website](#)

[LinkedIn](#)

## SKILLS

- Languages: C++ (advanced), Python (advanced), Fortran (intermediate)
- HPC: OpenMP, TBB, MPI (mpi4py), SLURM
  - ↳ familiar with: SYCL, Thrust, Kokkos, CUDA
- Performance: Profiling (perf, Malt, Maqao), Optimisation, Vectorisation
- Tools: Git, CMake, Docker, Linux, CI/CD, unit testing.
- Data Analysis: Scipy, Pandas, sklearn, PyTorch, JAX

## SOFT SKILLS

- Rigor and work organisation
- Autonomy and adaptability
- Written and oral communication
- Ability to explain complex technical concepts
- Technical ownership and sense of responsibility

## LANGUAGES

- French (native)
- English (professional)
- Japanese (beginner)

## HOBBIES

- Sports: Boxing, badminton
- Programming, video games, motorcycling, hiking

# TOM LACLAVÈRE

Doctor in Astrophysics · Software Engineer - Numerical Simulation & HPC

## EXPERIENCE

2023-2026

CNRS |  
Université  
Paris-Cité |  
APC Laboratory

### PhD - QUBIC Experiment

- Lead developer of the simulation, reconstruction and data analysis pipeline ([qubicsoft](#), > 30k LoC, > 500 commits).
- Large-scale linear system solving: numerical methods & machine learning.
- Parallelisation: mpi4py (Python), OpenMP (Fortran).
- Deployment on SLURM cluster (up to 20 nodes, 100-200 GB of data processed per run).
- Development of calibration and noise analysis tools for data validation.

2025-

GitHub  
Project

### HPC / Simulation Project - 3DPhysicsEngine (C++23)

- 3D rigid body simulation engine: custom math library, two-phase collision pipeline, three integrators (Euler, Verlet, RK4)
- Benchmarks: 150+ configurations (solver × time step), convergence, performance and energy drift quantified
- Current phase: CPU optimization (SIMD, OpenMP); roadmap: CUDA/SYCL, MPI

## EDUCATION

2023-2026

### PhD in Fundamental Physics - Astrophysique

- QUBIC Data Analysis: Realistic astrophysical components reconstruction and atmospheric mitigation using spectral imaging.(APC)

2025

### Gray Scott Summer School - HPC (CC-FR/LAPP)

- TBB, OpenMP, OpenMPI, SYCL, CUDA, Thrust
- C++, Fortran, Rust, Python, Julia

2023

### **Master Nuclei, Particles, Astroparticles & Cosmology (NPAC)** - Mention bien (Université Paris-Cité)

2021

### **Bachelor's in Fundamental Physic** - Mention Très bien (Université Paris-Cité)

2018-2020

### **CPGE, Physics-Chemistry** - Admitted to Centrale (Cité Scolaire Bertran-de-Born)

## TEACHING

2024 - 2026

CentraleSupélec  
Université  
Paris-Cité  
Ecole  
d'Ingénieur  
Denis Diderot

### Adjunct Teaching Assistant

- Cosmology - 1ère année - Tutorials (35h) - 2026
- Physique Numérique - Master 1 - Tutorials (36h) - 2024 & 2025
- Champs Electromagnétique - 1ère année - Tutorials (24h) - 2024
- Bruits - 2ème année - Lab sessions (24h) - 2024